IntelliTwin: Asymmetric, Condition-Adaptive Conformal Prediction for Industrial Turbofan Predictive Maintenance

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Abstract—Predictive maintenance for industrial gas turbine engines requires timely, risk-calibrated replacement decisions to prevent catastrophic in-flight failures while minimizing premature overhaul costs. Conventional Remaining Useful Life (RUL) prognostic systems rely on point regression models that optimize cycle-level accuracy metrics without quantifying predictive uncertainty or aligning with asymmetric maintenance cost structures.

In this work, we present IntelliTwin, a framework integrating data-driven base predictors (Gradient Boosted Regressors and Long Short-Term Memory networks) with hierarchical, decision-aware conformal prediction. IntelliTwin enforces strict engine-level disjoint data partitioning, causal temporal windowing ($w = 30$, stride 1), equal engine weighting across all evaluations, and condition-stratified conformal calibration. By incorporating an asymmetric loss parameter ($\beta = -\ln 2$), the framework modulates nonconformity scores to reflect the severe cost penalties of component failure relative to preventive replacement.

We formalize an experimental evaluation protocol across NASA C-MAPSS sub-datasets FD001, FD002, and FD004, comparing calibrated lower bounds against four distinct matched-conservatism baselines across three families. Statistical significance is evaluated via paired engine-cluster bootstrap inference ($B = 10{,}000$) with pre-registered practical-effect margins and Bonferroni family-wise error rate control (98.75% confirmatory confidence intervals).

Provisional Status Statement: This document represents the pre-registered methodology and evaluation structure; empirical evaluation on the held-out test splits remains pending experimental execution under stable update gates UPD-02, UPD-03, and UPD-04.

Index Terms—Predictive Maintenance, Conformal Prediction, Remaining Useful Life, Turbofan Engines, Digital Twin, Asymmetric Loss, C-MAPSS.

This research draft reflects pre-registered methodology and protocol specifications frozen under commit 224c741f5f3da19821a708e0a6da417d4810e9da. Held-out experimental results are blinded and pending completion of experimental verification gates (UPD-02, UPD-03, UPD-04).

I. Introduction

In safety-critical cyber-physical systems, such as commercial aviation turbofan engines, unexpected component failure incurs catastrophic operational, financial, and life-safety hazards. Consequently, industrial fleet operators have historically adhered to rigid preventive maintenance schedules, overhauling major subsystems at fixed operational intervals (flight hours or cycles). While preventive scheduling mitigates catastrophic in-flight failure, it imposes severe economic costs by routinely discarding components with substantial remaining physical life [1], [2].

Predictive Maintenance (PdM) leverages onboard multi-sensor telemetry to transition from time-based maintenance to condition-based, opportunistic intervention. Machine learning prognostics aims to estimate the Remaining Useful Life (RUL) of degrading machinery, enabling timely maintenance before failure occurs. In recent years, deep neural architectures—including Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) recurrent networks, and gradient-boosted decision trees—have achieved remarkable point-prediction accuracy on standardized benchmarks such as NASA C-MAPSS.

However, deterministic point predictors $\hat{y}$ provide no quantification of predictive uncertainty. In sequential decision environments, a deterministic forecast is fundamentally incomplete: an operator cannot determine whether a predicted life of $\hat{y} = 15$ cycles carries high confidence or a substantial probability of immediate breakdown at cycle 2. Furthermore, standard models are trained under symmetric loss functions (such as Mean Squared Error), which penalize early and late errors equally. In physical operations, this symmetry is catastrophic: predicting that an engine will survive longer than it actually does ($d > 0$) risks uncontained in-service failure, whereas predicting failure prematurely ($d < 0$) merely incurs early mainte-

nance cost [3], [4].

To provide rigorous probabilistic safety bounds, prognostic models must emit calibrated prediction intervals $[L(x), U(x)]$ guaranteeing nominal coverage $1 - \alpha$. While heuristic probabilistic methods (e.g., Monte Carlo dropout and Gaussian processes) rely on restrictive parametric assumptions, Conformal Prediction [5] provides distribution-free, finite-sample marginal coverage guarantees under data exchangeability.

Despite this theoretical appeal, applying conformal prediction to industrial asset prognostics reveals critical research challenges:

1) Autocorrelated Degradation Trajectories: Telemetry cycles within an engine violate observation exchangeability. Standard pooled conformal prediction over-weights long-lived engines, introducing survivor bias. Hierarchical, engine-weighted quantile computation [6] is required to restore exchangeability at the independent engine cluster level.
2) Subgroup Reliability Breakdown: Global conformal calibrators guarantee marginal coverage averaged across all operating cycles, but frequently exhibit acute localized undercoverage during high-stress flight regimes or during accelerated terminal wear stages [7], [8].
3) The Downstream Decision Gap: Downstream sequential maintenance policies trigger replacement actions when the lower prediction bound drops below an operational threshold ($L(t) \leq \tau$). Conformal calibration methods that treat intervals symmetrically or arbitrarily expand interval width can induce extreme operational conservatism, scrapping components prematurely under the guise of high coverage.

To bridge this gap, this paper investigates decision-aware, condition-adaptive conformal calibration for turbofan RUL prognostics. We present an end-to-end framework integrating causal temporal windowing, frozen point-predictor benchmarks, exact rational hierarchical quantiles, and asymmetric nonconformity score inversion, evaluated within a discrete-event sequential maintenance simulator.

II. Related Work

A. Remaining Useful Life Point Prognostics

Prognostic health management of aero-engines was popularized by the NASA C-MAPSS dataset release [1], which formalized run-to-failure simulation across varied operational flight conditions and fault modes. Standard point predictors model RUL using deep neural networks or gradient-boosted decision trees. While the C-MAPSS asymmetric scoring metric heavily penalizes overestimation ($d > 0$), base models are routinely optimized under symmetric loss objectives. Furthermore, point predictions provide no confidence intervals, depriving operators of actionable risk metrics.

B. Conformal Uncertainty Quantification in PHM

Conformal prediction [5] has emerged as an authoritative distribution-free uncertainty framework. Conformalized Quantile Regression (CQR) was adapted to aero-engines by Diao et al. [4] using LSTM networks on C-MAPSS FD001 and FD002, and by Robinson [9] across all four C-MAPSS subsets. However, standard split conformal methods assume observation exchangeability. Because sensor telemetry is autocorrelated within an engine, pooling raw cycles violates exchangeability and over-represents long-lived engines. Lee, Barber, and Willett [6] proved that for clustered hierarchical data, finite empirical mass must be bounded by $\frac{m}{m+1}$, with the remaining $\frac{1}{m+1}$ probability mass placed at $+\infty$, using engine-weighted quantiles to guarantee finite-sample coverage for unseen clusters.

C. Conditional Reliability Diagnostics and Maintenance Economics

Recent studies emphasize that marginal coverage obscures severe conditional vulnerabilities. Yang et al. [7] audited empirical calibration across operating-regime shifts in rotating bearings, revealing acute localized undercoverage. BenAbdennour [8] demonstrated grouped validation and interval-safety calibration across degradation domains, showing non-linear expansion of uncertainty during terminal wear stages.

Concurrently, uncertainty-aware maintenance scheduling has been investigated via simulation and reinforcement learning [3], [10], [11]. Cortes-Gomez et al. [12] introduced Utility-Directed Conformal Prediction (UDCP), demonstrating that tailoring nonconformity scores to downstream decision utility substantially reduces operational loss. IntelliTwin synthesizes these domains, evaluating whether asymmetric decision-aware score inversion can improve sequential maintenance utility while preserving subgroup reliability and coverage safety.

III. Problem Formulation and Hypotheses

A. Trajectory Formulation and Dual-Horizon Targets

Let $i \in \{1, \ldots, N\}$ denote an engine unit, and let $t \in \{1, \ldots, T_i\}$ denote operational flight cycles, where T_i is the cycle of total failure. At each cycle t, an engine produces raw telemetry $x_{it} \in \mathbb{R}^{17}$ comprising 3 operating settings and 14 qualified degradation sensors.

Causal sliding windows of length $W = 30$ cycles with stride 1 and strictly no padding are extracted:

$$X_{it} = [x_{i,t-29},\ x_{i,t-28},\ \ldots,\ x_{it}] \in \mathbb{R}^{30\times 17} \tag{1}$$

Qualified inference endpoints are $V_i = \{30, 31, \ldots, T_i\}$, yielding $n_i = T_i - 29$ valid predictions.

Under ADR-012, two distinct target horizons are formalized:

$$y_{\text{train}}(t) = \min(125,\ T_i - t) \quad \text{(Capped Training Target)} \tag{2}$$

$$y_{\text{true}}(t) = T_i - t \quad \text{(Uncapped Evaluation \& Calibration Target)} \tag{3}$$

Base models are optimized on y_{train} to prevent overfitting on early-life sensor plateaus. Calibration nonconformity scores and downstream maintenance simulations evaluate true uncapped longevity y_{true}.

B. Conformal Prediction Intervals

Given a base predictor point estimate $p_{it} = \max(0.0,\ \hat{y}(X_{it}))$, a conformal calibrator constructs a prediction interval:

$$C(X_{it}) = [L_{it},\ U_{it}] \tag{4}$$

such that empirical marginal coverage satisfies $\mathbb{P}(y_{\text{true}} \in C(X)) \geq 1 - \alpha$. In this study, nominal coverage is set to 90% ($\alpha = 0.10$), with secondary sensitivities at 80% and 95%.

C. The Pre-Registered Hypothesis Framework (ADR-018)

The study evaluates four pre-registered hypotheses governed by the following decision logic:

- RQ1 (Maintenance Utility):

  $H_{0,1} : \Delta_{\text{cost}} \leq 0.20$ SCU or $E_{\text{coverage}} \geq 0.05$ or $\Delta_{\text{fail}} > 0.0$

  $H_{1,1} : \Delta_{\text{cost}} > 0.20$ SCU and $E_{\text{coverage}} < 0.05$ and $\Delta_{\text{fail}} \leq 0.0$

  evaluated against global_hcp and regime_hcp under primary_reference on FD002.
- RQ2 (Subgroup Reliability):

  $H_{0,2} : \Delta_{\text{error}} \leq 0.01$ or $D_{\text{width}} \geq 1.0$ or $D_{\text{score}} \geq 1.0$

  $H_{1,2} : \Delta_{\text{error}} > 0.01$ and $D_{\text{width}} < 1.0$ and $D_{\text{score}} < 1.0$

  evaluated across operating regimes (C_1–C_6) and lifecycle stages (Early/Mid/Late).
- RQ3 (Decision-Value Attribution):

  $$H_{1,3} : \Delta_{\text{cost}} > 0.20\,\text{SCU over all four prespecified matched controls} \tag{5}$$
- RQ4 (Generalization & Non-Harm):

  $$H_{1,4} : \Delta^{\text{agg}}_{\text{cost}} > 0.20\,\text{SCU},\ \Delta^{\text{agg}}_{\text{error}} > 0.01,\ \text{and no cell harm} \tag{6}$$

  evaluated across the balanced 6-cell matrix (FD001, FD002, FD004 × GBR, LSTM).

IV. Proposed Methodology

A. Data Pipeline, Preprocessing, and Leakage Controls

The empirical pipeline processes C-MAPSS subsets FD001, FD002, and FD004 (totaling 609 engines). All data partitioning is executed strictly at the independent engine cluster level: approximately 50% training, 20% calibration, 10% validation, and 20% held-out test engines (manifests preserved in data/splits/).

A locked 17-feature schema is enforced under ADR-013: 3 operational flight settings (altitude, Mach number, throttle resolver angle) and 14 informative degradation sensors (pruning 7 sensors exhibiting zero or near-zero variance). All preprocessor scaling parameters (medians and interquartile ranges) and K-Means operational regime cluster centroids ($K = 6$) are fit strictly on the training partition; calibration, validation, and test engines are transformed using frozen training scalers.

Causal temporal windowing extracts chronological sequences of length $W = 30$ cycles with stride 1 and strictly no padding. To ensure fair architectural comparison under an identical temporal information horizon, the LSTM receives sequential tensors of shape $(30, 17)$, while the GBR receives flattened 510-dimensional vectors (30×17).

B. Frozen Base Predictor Matrix

Under ADR-019 and ADR-020, two point-predictor model families were optimized on validation data across seeds 2026–2027 and trained across five official research seeds (2026, 2027, 2028, 2029, 2030):

- Histogram Gradient Boosting Regressor (GBR): Scikit-learn HistGradientBoostingRegressor operating on flattened 510-dim vectors. Winning configurations: gbr_lr010_leaf15_l2_0 for FD001 (RMSE 28.42 cycles), and gbr_lr010_leaf31_l2_0 for FD002 (RMSE 34.66 cycles) and FD004 (RMSE 48.83 cycles). Execution is fully deterministic across seeds ($\sigma_{\text{RMSE}} = 0.0000$).
- Unidirectional LSTM: PyTorch LSTM ($H = 64$, dropout 0.20, learning rate $\eta = 0.0010$) with early stopping on validation uncapped RMSE. Winning configuration: lstm_h64_lr0010 across all datasets (Validation RMSE: FD001 = 30.08, FD002 = 35.25, FD004 = 48.25 cycles; seed standard deviation $\sigma \leq 0.36$ cycles).

All 30 checkpoints are hashed and immutably indexed in experiments/base_models/model_registry.json.

C. Hierarchical Engine-Weighted Conformal Calibration

Standard pooled conformal prediction introduces survivor bias in industrial run-to-failure telemetry because long-lived engines contribute more sliding windows than short-lived units. To establish finite-sample guarantees for unseen engines, IntelliTwin implements hierarchical engine-weighted quantiles following Lee, Barber, and Willett [6].

Let m denote the number of calibration engines contributing valid endpoints, where engine i contains $n_i = T_i - 29$ valid endpoints. Each individual endpoint (i, t) receives an exact mass:

$$w_{it} = \frac{1}{m \cdot n_i} \tag{7}$$

ensuring that every engine contributes exactly $\frac{1}{m}$ of total finite weight. Total finite probability mass is bounded by $\frac{m}{m+1}$, with the remaining $\frac{1}{m+1}$ probability mass assigned to $+\infty$:

$$F_{\text{hierarchical}}(q) = \frac{1}{m+1} \sum_{i=1}^{m} \frac{1}{n_i} \sum_{t=1}^{n_i} \mathbb{I}(s_{it} \leq q) \tag{8}$$

The conformal threshold q^* for nominal coverage $1 - \alpha$ is the smallest score satisfying $F_{\text{hierarchical}}(q^*) \geq 1 - \alpha$.

If no finite score satisfies this condition, $q^* = +\infty$. Quantiles are computed using exact rational arithmetic (fractions.Fraction).

D. Formulations of Evaluated Conformal Methods

Let $p \geq 0$ denote the base point prediction, $y = y_{\text{true}} \geq 0$ denote uncapped ground-truth RUL, and $g \in \{C_1, \ldots, C_6\}$ denote the observed operational regime.

1) Global Hierarchical Conformal Prediction (global_hcp): Uses symmetric absolute residuals:

$$s(p, y) = |y - p| \tag{9}$$

A single global threshold q_{global} is fitted, yielding:

$$L = \max(0,\ p - q_{\text{global}}), \quad U = p + q_{\text{global}} \tag{10}$$

2) Condition-Aware Hierarchical Conformal Prediction (regime_hcp): Uses symmetric absolute residuals, fitting separate regime-specific thresholds q_g for each operating regime g:

$$L = \max(0,\ p - q_g), \quad U = p + q_g \tag{11}$$

If regime g contains fewer than $m_g = \max(10, \lceil 1/\alpha \rceil - 1)$ calibration engines, execution falls back automatically to q_{global} with logging.

3) Proposed Decision-Aware Asymmetric Conformal Prediction (decision_aware_hcp): Modulates interval width and asymmetry near the critical end-of-life decision boundary:

$$\sigma(p) = \left(1 + \frac{p}{125}\right)^{\gamma}, \quad z(p) = \exp\left(-\frac{p}{20}\right) \tag{12}$$

$$a(p) = \sigma(p)\exp(\beta z(p)), \quad b(p) = \sigma(p)\exp(-\beta z(p)) \tag{13}$$

where $\gamma \in \{0, 1, 2\}$ is the scaling exponent and $\beta = \beta_{\text{code}} \ln(2)$ with $\beta_{\text{code}} \in \{-1, 0, 1\}$ governs asymmetry. The nonconformity score and inverted interval are:

$$\text{score}(p, y) = \max\left(\frac{p - y}{a(p)},\ \frac{y - p}{b(p)},\ 0\right) \tag{14}$$

$$L = \max\left(0,\ p - q_g \cdot a(p)\right), \quad U = p + q_g \cdot b(p) \tag{15}$$

E. Mathematical Mechanics of Negative Beta ($\beta < 0$)

When $\beta_{\text{code}} = -1$ ($\beta = -\ln 2 \approx -0.693147$), as $p \to 0$ near end of life ($z(p) \to 1$), $a(p)$ is halved ($\sigma(p) \cdot 0.5$) while $b(p)$ is doubled ($\sigma(p) \cdot 2.0$).

Critical Mathematical Fact: Holding p, q, and γ fixed, negative β decreases the lower-side scale $a(p)$ and increases the upper-side scale $b(p)$, which causes the unclipped lower bound $p - q_g \cdot a(p)$ to rise (or remain at zero after clipping). This mathematical fact does not justify claiming earlier replacement or guaranteed failure protection; refitted empirical nonconformity thresholds q^* differ between candidates, and true operational utility must be measured directly through sequential maintenance simulation.

In leave-one-engine-out validation selection across seeds 2026–2030, candidate gamma2_betam1 ($\gamma = 2, \beta_{\text{code}} = -1$) was selected as the optimal feasible configuration across all six mandatory cells and frozen under ADR-021.

V. Experimental Setup

A. Benchmark Datasets and Engine-Level Partitions

We evaluate IntelliTwin on three benchmark subdatasets from the NASA C-MAPSS turbofan engine degradation simulation suite [1]:

1) FD001: Single sea-level operating condition, single failure mode (high-pressure compressor degradation), comprising 100 run-to-failure engine trajectories.
2) FD002: Six diverse operating flight conditions, single failure mode (HPC degradation), comprising 260 engine trajectories.
3) FD004: Six diverse operating flight conditions, two concurrent failure modes (HPC and fan degradation), comprising 249 engine trajectories.

To prevent temporal and engine-level data leakage, partitioning is strictly executed at the engine level using a fixed master seed (2026) prior to any data preprocessing (50% training, 20% calibration, 10% validation, 20% held-out test):

- FD001 (100 engines): 50 training (50%), 20 calibration (20%), 10 validation (10%), and 20 held-out test engines (20%).
- FD002 (260 engines): 130 training (50%), 52 calibration (20%), 26 validation (10%), and 52 held-out test engines (20%), stratified across engine lifetime quartiles to balance short, medium, and long-life engine profiles across splits.
- FD004 (249 engines): 124 training (50%), 50 calibration (20%), 25 validation (10%), and 50 held-out test engines (20%).

Across all three datasets, 304 engines are allocated to training, 122 to calibration, 61 to validation, and 122 to final held-out testing (total 609 engines).

B. Feature Space and Causal Windowing

The raw engine streams comprise 3 operational settings and 21 sensor measurements per operational cycle. In accordance with training-only qualification protocols, seven sensor channels showing zero or near-zero variance across operating cycles are excluded, yielding a standardized 17-dimensional feature vector $\mathbf{x}_t \in \mathbb{R}^{17}$ (3 operational settings and 14 informative sensor channels). Robust min-max scaling is fitted strictly on training engines.

Temporal trajectories are windowed with an exact causal horizon of $w = 30$ cycles, unit stride ($s = 1$), and strict avoidance of zero-padding. Consequently, model predictions initiate exclusively at cycle $t = 30$. In training splits, target RUL is capped at $R_{\max} = 125$ cycles to reflect early healthy operational regimes where degradation is physically negligible. In calibration, validation, and held-out test splits, RUL is preserved uncapped to evaluate true physical end-of-life trajectories.

C. Base Degradation Predictors

We investigate two standard, distinct algorithmic paradigms operating on identical 30-cycle causal information horizons:

1) Gradient Boosted Regressor (GBR): LightGBM ensemble operating on flattened $30 \times 17 = 510$-dimensional causal feature vectors.
2) Long Short-Term Memory (LSTM): Recurrent neural network architecture processing sequential 30×17 temporal tensors.

This yields a 3×2 experimental matrix (6 experimental cells): FD001-GBR, FD001-LSTM, FD002-GBR, FD002-LSTM, FD004-GBR, and FD004-LSTM. To capture training stochasticity, each cell is trained across five random seeds $(2026, 2027, 2028, 2029, 2030)$, producing 30 frozen base model checkpoints in the registry.

D. Calibration Variants and Control Baselines

On the reserved calibration engine splits, we evaluate three conformal calibration variants:

- Global HCP (global_hcp): Standard split-conformal calibration with engine-weighted empirical quantiles.
- Regime-Stratified HCP (regime_hcp): Cluster-conditional nonconformity scaling and quantile fitting conditioned on operating regime $k \in \{1, \ldots, 6\}$.
- Decision-Aware HCP (decision_aware_hcp): Asymmetric nonconformity scaling parameterized by risk aversion parameter $\beta = -\ln 2$, penalizing RUL overestimation.

To rigorously address RQ3, we evaluate four named controls across three distinct baseline families:

1) Heuristic Interval Family:
   - Matched-Width Interval: Symmetric heuristic interval centered at $\hat{y}_t$ with constant half-width matched to the mean half-width of the calibrated HCP interval.
   - Widened Interval: Heuristic interval expanded by $1.25\times$ the matched half-width.
2) Point Safety Margin Family:
   - Point Safety Margin: Point prediction trigger shifted downward by a fixed margin $\Delta = 10$ cycles: replace when $\hat{y}_t \leq \tau + \Delta$.
3) Earlier Fixed Policy Family:
   - Earlier Intervention: Direct threshold inflation: replace when $\hat{y}_t \leq \tau_{\text{early}} = \tau + 15$ cycles.

E. Maintenance Simulation and Cost Scenarios

Sequential policy execution is evaluated on unseen test engine trajectories using the discrete-event simulator. The primary economic scenario specifies $C_{\text{preventive}} = 10$, $C_{\text{unplanned}} = 100$ (cost ratio $10:1$), and lead-time $L_{\text{lead}} = 2$ cycles. Robustness is assessed across sensitivity scenarios spanning cost ratios $5:1$ ($C_{\text{unplanned}} = 50$), $20:1$ ($C_{\text{unplanned}} = 200$), and varying lead-time horizons ($L_{\text{lead}} \in \{0, 5\}$).

F. Statistical Inference and Decision Protocol

To evaluate confirmatory hypotheses H1–H4 without distributional assumptions, we apply paired engine-level cluster bootstrap inference:

- Resampling Unit: Complete engine trajectories are resampled with replacement ($B = 10{,}000$ resamples) preserving internal temporal correlations.
- Weighting: Equal weighting per engine is applied across all metrics.
- Missing-Group Bound Propagation: If a bootstrap sample lacks representation for a specific operational cluster, worst-case bounds ($[0, 1]$ for coverage, $[0, \infty)$ for width) are propagated.
- Family-Wise Error Rate: Confirmatory decision confidence intervals are computed at 98.75% two-sided level ($\alpha_{\text{conf}} = 0.0125$) implementing Bonferroni correction across the four primary research questions.
- Intersection-Union Test (IUT): To decide Supported, Unsupported, or Inconclusive, the lower/upper bounds of the confidence intervals must strictly satisfy the pre-registered practical margin thresholds $\delta_{\text{cov}} = 0.00$, $\delta_{\text{cost}} = 0.05$, and safety constraint $\Delta_{\text{fail}} \leq 0$.

VI. Results and Confirmatory Evaluation

Status Notice: The empirical evaluations reported in this section represent the pre-registered reporting templates for the confirmatory evaluation. In accordance with the project's evidence protocol, all held-out test split evaluations remain blinded during active software development and verification. Empirical entries are designated PENDING awaiting completion of experimental gates UPD-02, UPD-03, and UPD-04. No synthetic or unverified empirical numbers are reported.

A. Base Degradation Prediction Performance

Point predictive performance is evaluated across the six experimental cells on the held-out test engine splits. Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) are computed with equal engine weighting across all test engines. Table I outlines the reporting structure.

TABLE I
Point Prediction Accuracy on Held-Out Test Engines (Mean ± Std across 5 Seeds)

Dataset	Architecture	RMSE (cycles)	MAE (cycles)
FD001	GBR	PENDING (UPD-02)	PENDING (UPD-02)
FD001	LSTM	PENDING (UPD-02)	PENDING (UPD-02)
FD002	GBR	PENDING (UPD-02)	PENDING (UPD-02)
FD002	LSTM	PENDING (UPD-02)	PENDING (UPD-02)
FD004	GBR	PENDING (UPD-02)	PENDING (UPD-02)
FD004	LSTM	PENDING (UPD-02)	PENDING (UPD-02)

B. Marginal and Conditional Interval Calibration

Conformal prediction intervals are evaluated at nominal confidence level $1 - \alpha = 0.90$ across methods: uncalibrated baseline (Gaussian heuristic), Global HCP, Regime-Stratified HCP, and Decision-Aware HCP. Metrics comprise engine-weighted empirical coverage (Cov), mean interval width ($\bar{W}$), and Winkler interval score ($\mathrm{IS}_{0.10}$). Table II summarizes the evaluation grid.

C. Sequential Maintenance Decision Outcomes

Downstream maintenance outcomes under the primary economic scenario ($C_{\text{preventive}} = 10$, $C_{\text{unplanned}} = 100$, lead-time $L_{\text{lead}} = 2$) are reported in Table III. Metrics include mean maintenance cost per engine ($\bar{C}$), unplanned failure proportion (P_{fail}), and mean wasted operational life ($\bar{U}$).

D. Hypothesis Decision Outcomes (RQ1–RQ4)

Confirmatory statistical inference is conducted via 10,000 paired engine-level cluster bootstrap resamples. Table IV presents the formal decision matrix with Bonferroni-corrected 98.75% confidence intervals.

VII. Discussion

A. Methodological Implications for Turbofan Prognostics

The core inquiry of the IntelliTwin framework is whether distribution-free uncertainty quantification provides measurable, operational value to industrial decision-makers beyond standard point prognostics or ad-hoc safety margins. Traditional prognostic paradigms focus almost exclusively on minimizing cycle-level error metrics (e.g., RMSE or asymmetric scoring functions). However, in mission-critical applications such as aviation gas turbine maintenance, cycle-level accuracy does not translate directly into operational cost efficiency. A maintenance policy driven by an uncalibrated point predictor is structurally vulnerable to overestimation errors, where an engine is allowed to operate beyond its physical failure boundary, incurring catastrophic unplanned replacement penalties.

By establishing finite-sample marginal coverage guarantees via split-conformal inference, IntelliTwin provides decision-makers with a mathematically principled lower bound $L_t(\alpha)$. Nevertheless, conformal guarantees hold marginally across the calibration exchangeability distribution. In sequential degradation processes, temporal autocorrelation and operating regime transitions can cause local conditional coverage to deviate significantly from the nominal $1 - \alpha$ rate.

B. Conservatism, Asymmetric Penalties, and Negative β

A critical methodological contribution of this work is the formalization of asymmetric, decision-aware conformal calibration. In maintenance scheduling, the cost asymmetry between unplanned catastrophic failure ($C_{\text{unplanned}} = 100$) and scheduled preventive overhaul ($C_{\text{preventive}} = 10$) dictates that underestimating RUL (which leads to premature replacement) is substantially less hazardous than overestimating RUL (which leads to in-flight shutdown or catastrophic component damage).

The asymmetry parameter $\beta = -\ln 2 \approx -0.6931$ modulates the nonconformity score scaling factors:

$$a(p_t) = \exp(-\beta \cdot p_t), \quad b(p_t) = \exp(\beta \cdot p_t). \tag{16}$$

For negative β, as failure probability p_t increases, $a(p_t)$ decreases below 1, while $b(p_t)$ increases above 1. At fixed nonconformity thresholds, this asymmetric scaling contracts the lower distance and expands the upper distance, shifting the unclipped lower bound L_t upward. However, as established in the project's statistical protocol, negative β does not mathematically guarantee earlier maintenance across all operational trajectories. Because the calibration set quantiles q^* are refitted on asymmetrically scaled nonconformity scores, the empirical distribution of scores changes. Consequently, the actual operational effect of asymmetric calibration must be validated through sequential simulation rather than theoretical assumption.

C. Regime Stratification and Subgroup Reliability

In complex operational environments such as C-MAPSS FD002 and FD004, turbofans traverse six distinct operating flight regimes characterized by substantial variations in altitude (0–40,000 ft), Mach number (0–0.84), and throttle resolver angle. Global conformal calibration aggregates residuals across all regimes, which can mask severe local undercoverage in rare or highly stressed operating conditions.

Regime-stratified HCP explicitly conditions nonconformity scores on the active flight regime cluster $k \in \{1, \ldots, 6\}$. While this guarantees conditional coverage asymptotically as calibration sample sizes per regime grow large, finite-sample calibration across rare regimes introduces higher quantile variance. When a rare regime has fewer than the required sample threshold ($N_k < 100$), the framework executes a deterministic fallback to the pooled global nonconformity quantile, ensuring that the resulting prediction intervals remain well-defined and conservative.

D. Interpreting Potential Hypothesis Outcomes

The pre-registered statistical protocol explicitly defines decision rules for three potential outcomes across all four hypotheses:

- Supported: The lower confidence bound of the contrast metric strictly exceeds the pre-registered practical margin, demonstrating statistically significant and practically meaningful superiority.
- Inconclusive: The confidence interval spans the decision boundary, or statistical power is insufficient to confirm or refute the practical margin. In this case, no definitive superiority claim is made, and the result is cataloged as requiring expanded engine sample sizes.

TABLE II
Interval Calibration Performance at Nominal 90% Confidence Level on Held-Out Test Splits

Dataset	Base Model	Calibration Method	Empirical Cov. (%)	Mean Width (cycles)	Winkler Score
FD001	GBR	Uncalibrated Baseline	PENDING	PENDING	PENDING
FD001	GBR	Global HCP	PENDING	PENDING	PENDING
FD001	GBR	Decision-Aware HCP	PENDING	PENDING	PENDING
FD002	GBR	Global HCP	PENDING	PENDING	PENDING
FD002	GBR	Regime HCP	PENDING	PENDING	PENDING
FD002	GBR	Decision-Aware HCP	PENDING	PENDING	PENDING
FD002	LSTM	Global HCP	PENDING	PENDING	PENDING
FD002	LSTM	Regime HCP	PENDING	PENDING	PENDING
FD002	LSTM	Decision-Aware HCP	PENDING	PENDING	PENDING
FD004	LSTM	Global HCP	PENDING	PENDING	PENDING
FD004	LSTM	Regime HCP	PENDING	PENDING	PENDING
FD004	LSTM	Decision-Aware HCP	PENDING	PENDING	PENDING

TABLE III
Sequential Maintenance Policy Outcomes on Held-Out Test Engines ($C_{\text{prev}} = 10, C_{\text{unplanned}} = 100, L_{\text{lead}} = 2$)

Dataset	Model	Policy Strategy	Cost / Engine	Failures (%)	Wasted Life (cycles)
FD002	GBR	Uncalibrated Point Policy	PENDING (UPD-03)	PENDING	PENDING
FD002	GBR	Point Safety Margin (+10)	PENDING (UPD-03)	PENDING	PENDING
FD002	GBR	Matched-Width Interval	PENDING (UPD-03)	PENDING	PENDING
FD002	GBR	Widened Interval (1.25×)	PENDING (UPD-03)	PENDING	PENDING
FD002	GBR	Earlier Intervention (+15)	PENDING (UPD-03)	PENDING	PENDING
FD002	GBR	Global HCP Lower Bound	PENDING (UPD-03)	PENDING	PENDING
FD002	GBR	Regime HCP Lower Bound	PENDING (UPD-03)	PENDING	PENDING
FD002	GBR	Decision-Aware HCP Bound	PENDING (UPD-03)	PENDING	PENDING

TABLE IV
Pre-Registered Confirmatory Hypothesis Decisions and Statistical Contrasts

Hypothesis	Contrast Metric	Decision Rule / Threshold	Estimate	98.75% CI	Final Decision
H1 (Coverage)	$\text{Cov}(\text{HCP}) - 0.90$	$\text{CI}_{\text{lower}} \geq 0.00$	PENDING	PENDING	PENDING (UPD-04)
H2 (Cost)	$(\bar{C}_{\text{point}} - \bar{C}_{\text{HCP}})/\bar{C}_{\text{point}}$	$\text{CI}_{\text{lower}} \geq 0.05 \wedge \Delta_{\text{fail}} \leq 0$	PENDING	PENDING	PENDING (UPD-04)
H3 (Controls)	$\bar{C}_{\text{ctrl}} - \bar{C}_{\text{HCP}}$	$\text{CI}_{\text{lower}} > 0 \vee (\Delta C \geq 0 \wedge \Delta_{\text{fail}} \leq 0)$	PENDING	PENDING	PENDING (UPD-04)
H4 (Regimes)	$\text{Cov}_{\text{worst}}(\text{Regime}) - \text{Cov}_{\text{worst}}(\text{Global})$	$\text{CI}_{\text{lower}} > 0.00$	PENDING	PENDING	PENDING (UPD-04)

- Unsupported: The upper confidence bound falls below the required threshold, or the safety constraint is violated (e.g., failure rate increases). This outcome directly falsifies the hypothesis, indicating that conformal calibration does not provide the anticipated decision benefit over simpler baselines.

By enforcing strict pre-registered boundaries and reporting all comparisons transparently, the research framework eliminates publication bias and ensures that conclusions are dictated entirely by empirical evidence.

VIII. Conclusion and Future Work

A. Summary of Methodological Framework

In this investigation, we formulated and pre-registered the IntelliTwin prognostic and predictive maintenance framework. Designed to address the critical gap between cycle-level machine learning accuracy and downstream operational cost in industrial turbofan fleets, IntelliTwin couples gradient boosted and recurrent deep learning base predictors with hierarchical, decision-aware conformal calibration.

The framework enforces rigorous methodological safeguards to preserve empirical validity:

1) Strict Leakage Isolation: Complete engine-level disjoint partitioning (train, calibration, validation, held-out test) executed prior to feature scaling or causal windowing ($w = 30$, stride 1, no padding).
2) Equal Engine Weighting: Evaluation metrics and conformal quantiles are computed with equal weighting per engine trajectory, eliminating length bias inherent in pooled cycle-level evaluations.
3) Principled Baseline Controls: Evaluation against four named control strategies across three distinct families (matched-width heuristic intervals, widened intervals, point safety margins, and earlier fixed interventions) to isolate the true economic value of calibrated uncertainty.
4) Pre-Registered Statistical Inference: Family-wise error rate control via Bonferroni-corrected 98.75%

confidence intervals generated through 10,000 paired engine-level cluster bootstrap resamples.

B. Current Status and Verification Gates

At the current stage of project execution, the theoretical formulation, data pipelines, base model training protocols, and statistical evaluation machinery have been fully specified, audited, and frozen. Empirical results on the held-out test splits remain blinded and pending final experimental execution by the development team.

The planned progression toward publication comprises:

- UPD-01: Verification of pretest reproduction audits and environment fixes.
- UPD-02 & UPD-03: Batch execution of held-out prediction, interval calibration, and discrete-event maintenance simulations.
- UPD-04: Formal computation of cluster-bootstrap confidence intervals and execution of Intersection-Union Test decision rules for RQ1–RQ4.
- UPD-06 & UPD-07: Integration of the replay-based digital twin demonstrator and independent clean-environment reproducibility verification.

Upon completion of these verification gates, the resulting empirical evidence will be integrated directly into this document following the established 10-step protocol.

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